

From an Egg to Computation

A question-driven journey from food, life to information and computation

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Acknowledgement and AI-assisted development

This essay emerged through a dialogue between human questioning and AI-assisted exploration. The initial questions, observations, conceptual connections, and overall direction came from me; ChatGPT helped me explore these ideas, test analogies, identify distinctions, reorganize the argument, and turn the evolving discussion into a coherent essay.

The final text is therefore neither presented as purely AI-generated nor as written without AI assistance. It is the result of an iterative process in which human curiosity provided the questions and direction, while AI contributed discussion, refinement, and language-level assistance. I reviewed and shaped the resulting arguments and take responsibility for the final version.

Prefatory note: Why physics and information?

Since 2019, I have been a theoretical physics student. Over the past few years, particularly from 2025 onward, I gradually began to perceive physics and information science not as two separate subjects, but as two worlds whose boundaries were slowly diminishing. Each has its own language, questions, mathematical backbone, and ways of looking at reality. What interests me most is what happens when we try to sense a connection between them and explore the regimes where they overlap.

For me, entropy became one starting point for that connection. A quantity associated with uncertainty and information also appears at the heart of statistical mechanics and thermodynamics. It offers a bridge between how we describe a system and its physical behavior, though it is not the only connection between these fields.

Physics asks how nature works, what laws govern physical systems, and how their behavior can be measured and predicted. Information science brings another set of questions into view: how can information be represented, encoded, transformed, stored, communicated, compressed, and recovered? Where the two meet, these are not merely questions about abstract symbols. They concern what information can do through a physical system, and what the physical world allows us to do with it.

This is where one encounters classical and digital computing, continuous-variable computing, quantum computing, quantum communication and networks, neuromorphic computing, and many other ways of processing information. They may look like very different technologies, but underneath them lies a common question: what can we do with information, and what does the physical world allow us to do with it? Each approach brings its own physics, mathematics, possibilities, limitations, and problems.

Quantum computing is one example. The interest is not simply in building a faster computer, but in understanding how physical systems such as photons, superconducting circuits, atoms, and ions can be

controlled and used to encode information. Questions about noise, error correction and mitigation, fault-tolerant computation, and larger quantum networks connect this work to quantum matter, simulation, metrology and sensing, control theory, electronics, and manufacturing. My own interest is in developing skills and experience in quantum computing and error correction, with the goal of understanding how fault-tolerant systems can be realized.

Yet another layer draws me towards the foundations: why does nature allow quantum information to behave as it does? Why does entanglement exist, and what makes quantum information different from classical information? More broadly, what does it mean for information to exist in a physical universe? These questions reflect my interest in natural philosophy, alongside the practical work of learning how information can be manipulated and protected.

The information-science side opens an equally rich mathematical world: algorithms, computational complexity, coding and decoding, compression, memory, storage, and data recovery. Across this intersection, tools such as tensor networks, algebras, topology, computational physics, geometry, group theory, and Lie theory provide ways to study states, transformations, and symmetries. I find this mathematical territory less comfortable, but highly relevant to the research methods I want to develop. Mathematics gives the questions structure, while technology makes some of their consequences tangible.

Information gives us a new way to think about physical systems, while physics tells us what information can actually do. For me, becoming a better information physicist therefore means more than learning a particular set of techniques. It also means learning to recognize related questions when they appear in places that initially seem unrelated. The essay that follows is one attempt to explore that habit of thought, beginning with something much more ordinary than a laboratory or a computer.

Adapted from a personal reflection written in September 2026.

Introduction

The previous note described several ways in which information can be physically realized. Despite their different platforms, they invite a common question: *How does a physical system represent, transform, and preserve information, and what can it produce through that process?*

But what does information processing look like in nature? Perhaps we do not need to begin with a computer at all. We could start with something much older than any human technology, and much more ordinary in our daily lives: an egg. At first, the comparison may seem strange. An egg is not an engineered computer, and an embryo is not running a program in the conventional sense. Yet development gives us a concrete setting in which to examine how information, cellular machinery, resources, and environmental conditions work together.

Rather than asking whether an egg is a computer, we can ask: *Does a developing egg provide a working physical picture of information processing in nature?* More concretely, can it help us see how nature uses information to organize matter? This question is broader than any particular computing technology. It asks whether a naturally occurring process can make visible some of the roles we have been discussing in the language of physics and information science.

Before pursuing that connection, though, we need to ask a simpler question: what exactly is an egg? Or, in the most familiar setting, what exactly are we eating when we eat one?¹ There is a shell, some transparent white material, and a yellow yolk. We cook it, eat it, and move on. Yet a moment's pause opens up further questions: is the egg itself alive? Is the yolk alive? Is the white alive? Where, exactly, is the life in an egg? And if an egg can become a chicken, where is the chicken hidden before it becomes a chicken? Let us begin there and allow the connection to emerge one layer at a time.

¹Readers interested mainly in the dietary question may jump straight to the dietary side note. The author does not currently eat eggs; this began with wondering whether to start eating them again, and somehow ended up revisiting the connection between physics and information. Apparently, even breakfast can require a theoretical detour.

1 Is an egg alive?

Consider a typical chicken egg: a hard outer shell surrounds albumen, commonly called egg white, with the yolk at the center. Our first instinct might be to ask which of these parts is *alive*, but describing their contents does not immediately settle the question. The yolk is full of proteins, fats, vitamins, minerals, and other nutrients. The albumen contains mostly water and proteins. The shell consists primarily of calcium carbonate. None of these descriptions sounds like a living organism, yet the egg can, under the right circumstances, produce a chicken. So where is the life?

Perhaps the answer is the yolk, but that cannot quite be right. The yolk is mostly a nutritional reservoir. It contains enormous amounts of material and energy that a developing embryo can use, but most of the yolk itself is not an embryo. What about the egg white? The same distinction applies. Albumen provides water, proteins, and other resources, and also contributes to protection. But it is not itself the developing organism. So we have already encountered a strange situation:

A structure capable of becoming a living organism contains a great deal of material that is not itself the organism.

This suggests that we should distinguish the material from its organization. Once we make that distinction, the question becomes more precise: where is the actual beginning? The biologically important structure is associated with the oocyte, the female reproductive cell. On the surface of the yolk there is a small region called the germinal disc, associated with the cellular and nuclear machinery of the egg. The sperm, meanwhile, is itself a living reproductive cell. If sperm reaches the egg cell at the appropriate stage, fertilization occurs. The paternal and maternal genetic material combine, producing a zygote and initiating a new developmental trajectory.

What becomes visible here is not a miniature chicken hidden inside the yolk, waiting to grow larger. Instead, there is a living cellular system carrying information and machinery, surrounded by a large amount of stored material. The organism will be constructed progressively. This gives us our first conceptual distinction: living machinery $\neq$ material consumed by the machinery. The yolk, albumen, and shell are not the embryo. The embryo is the living, dynamically organized system that uses the surrounding material.

If the yolk is not the chicken, what is it doing there? The answer follows from a practical constraint: development costs resources. An embryo cannot simply walk to a refrigerator and find food, so it has to carry its developmental resources with it. So we might draw a very crude conceptual picture:

Developmental system + stored resources $\rightarrow$ growth and organization

There is an important subtlety in this picture. The embryo does not simply *turn the yolk into a chicken*. It takes molecules from the surrounding environment, processes them, rearranges them, and incorporates them into newly generated cells and tissues. The raw material is not the organization: amino acids are not muscles, lipids are not membranes in isolation, minerals are not bones in isolation, and water is not blood in isolation. The developing organism organizes matter, and this distinction will become increasingly important.

This brings us to fertilization. If the egg already exists before fertilization, what does fertilization actually add? It does not transform dead matter into living matter: the oocyte was already a living cell, and the sperm is also a living cell. So it is misleading to think:

non-life $\rightarrow$ fertilization $\rightarrow$ life.

A better description is:

existing living cells $\rightarrow$ fertilization $\rightarrow$ zygote $\rightarrow$ development $\rightarrow$ new organism.

Fertilization is therefore not the origin of life in the broadest sense, but a transition in the reproductive process. Together with the role of the yolk, this gives us another distinction: information and organization are not identical to matter and energy. The embryo needs matter and energy, but it also needs information and machinery capable of using those resources.

With that distinction in mind, we can ask what exactly is being carried from one generation to another. Consider the extraordinary scale difference. A chicken is a complicated multicellular organism. It contains bones, muscles, blood vessels, nerves, organs, feathers, and countless specialized cells. Yet its developmental process begins from a microscopic cellular system. Where, then, is all that complexity? It was not compressed physically into the cell. Rather, it contains information and machinery capable of generating the organization of a chicken. To see why this matters, consider a photograph: it contains information about a chicken's appearance, but it cannot build a chicken. A genetic system, by contrast, participates in a physical process that can produce the structure described by its developmental information. So perhaps the more interesting question is not:

Where is the chicken?

but:

Where is the mechanism that can produce the chicken?

That question shifts our attention from static objects to processes.

2 Information alone is not enough

The shift towards processes also reveals a limit to the idea of information alone. Suppose we somehow had the complete genetic sequence: would that sequence, sitting on a table, produce a chicken? Clearly not. Information requires a physical system capable of interpreting and acting upon it. DNA exists inside a complex cellular environment containing molecular machinery, enzymes, membranes, energy-transduction systems, signaling networks, and many other components. Therefore:

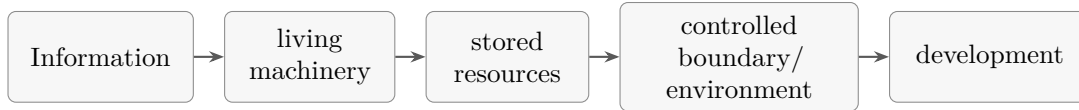
Information without machinery is not an operating process.

The embryo, then, is not simply DNA. It is a physical system in which information, molecular machinery, energy, matter, and environmental conditions interact continuously. We can therefore refine our picture: information contributes to development through its interaction with

- living machinery,
- resources,
- the environment.

Our picture is beginning to look less like a static object and more like a computational architecture, but we are not ready to call it a computer yet. There is still one piece to consider: the boundary. The embryo is not floating freely in an infinite environment. It exists inside a shell and a set of membranes that regulate its interaction with the outside world. The shell provides mechanical protection. It is not an impenetrable wall: gases can exchange through microscopic pores, while water loss and physical disturbances are constrained.

The boundary therefore does something more sophisticated than simply *protecting*: it controls interaction with the environment. This matters because a system cannot develop in complete isolation. It needs resources from its environment, yet unrestricted interaction can also disrupt it. A developmental system therefore needs something between complete isolation and complete exposure: a controlled boundary. Adding that boundary gives us a fuller picture:



Once these parts are distinguished, a more general pattern begins to emerge. Consider a seed: it too contains an embryo, stored nutrients, and protective structures. We find the same combination of embryonic machinery, information, stored resources, and a boundary. Like the egg, the seed does not contain a miniature version of the mature organism; it contains a system capable of becoming one. A human reproductive cell also brings together information, cellular machinery, and a developmental pathway. More generally, a biological cell has genetic information, molecular machinery, energy-processing systems, reservoirs of material, and a membrane controlling interaction with its environment. As the pattern returns, we begin to suspect that it describes something broader than reproduction: an architecture of organized transformation.

Could that architecture also help us think about computation? To explore the possibility, consider a very different object: a computer. It too requires several conceptually distinct ingredients:

- information to represent,
- rules or an algorithm describing transformations,
- physical machinery implementing those transformations,
- energy to operate the machinery,
- memory and other resources,
- an environment that must be controlled sufficiently for the computation to work,
- and an output produced after the transformation.

The resemblance to our biological picture is now easier to see. This does not make a chicken literally a computer in the engineering sense; rather, both systems can be described through a common abstraction:

A system receives or contains information, possesses machinery capable of transforming states, uses physical resources, interacts with an environment under constraints, and produces an organized result.

This is a much stronger analogy than simply saying:

“DNA is like computer code.”

DNA is only one component of the biological system. The analogy becomes more useful when we consider the whole architecture rather than treating a single component as the entire process.

3 Computation as a physical process

To make the comparison more concrete, imagine asking a computer to sort a list of numbers. The numbers are the information being represented, while an algorithm specifies how they should be compared and rearranged. Yet neither the list nor a written description of the algorithm performs the task by itself. A physical machine must carry out the operations, using memory, energy, and other resources. The result is an ordered list, but getting there requires more than knowing what the result should look like. It requires a process that can produce it.

This returns us to the distinction we found in the egg. A photograph of a chicken cannot build one, just as a description of a sorted list cannot do the sorting. In each case, we must distinguish information about a result from the machinery and transformations through which a result is reached. The comparison has limits, however. An embryo is not simply following a sorting algorithm, and its development is not a program written by an engineer. The phrase *natural information processing* directs our attention to how genetic information participates in cellular activity and development. In that sense, the developing egg offers a concrete example to examine, rather than only a visual metaphor. This does not establish it as a general-purpose computer or make it another technological platform alongside continuous-variable or neuromorphic computing.

The role of resources also becomes clearer in this example. Electrical energy allows the computer to operate, but it does not specify which list to sort or how to sort it. Similarly, the yolk provides energy-rich material without, by itself, determining what the embryo will become. Conversely, genetic information does not supply all the energy required for development. This is the sense in which we distinguish the two roles:

Information is not energy.

That distinction does not make information independent of matter. The numbers, instructions, and intermediate results must all have a physical representation. Rather, it separates what is represented and how it is transformed from the resources needed to carry out that transformation.

4 From material to organization

Taken together, these distinctions suggest that the deepest lesson from the egg is about organization rather than eggs alone. Imagine taking all the chemical substances present in an embryo and placing them randomly in a container. Would a chicken emerge? No, because the same atoms can exist in radically different organizations. A living organism is not merely a particular collection of molecules. It is a dynamically maintained organization of matter and information. The embryo continuously transforms resources while maintaining and creating structure. At this broad level, we can describe the developmental process as a sequence of transformations:

initial state $\rightarrow$ rules of transformation $\rightarrow$ resource consumption $\rightarrow$ interaction with environment
 $\rightarrow$ new state

For an engineered computer, we can make the information-processing steps more explicit:

represented input $\rightarrow$ specified operations $\rightarrow$ physical execution $\rightarrow$ intermediate states $\rightarrow$ represented output

These descriptions highlight a resemblance without establishing an equivalence. A process does not become a computation merely because it changes state or produces organization. In the computing example, we identify what the states represent and the operations that transform them. In the embryo, our comparison is more modest: it helps distinguish the different roles involved in development. With that limit in mind, we can put the correspondences side by side:

Biological development	Computing comparison
Genetic information	Information involved in transformations
Cellular machinery	Physical machinery implementing operations
Yolk and albumen	Resources supporting the process
Shell and membranes	Regulation of environmental interaction
Developmental dynamics	Evolution through successive states
Embryo	A system undergoing transformation
Mature organism	An organized outcome, rather than a literal computational output

But even this table should not be treated as a literal translation. The more fundamental correspondence is:

Information + machinery + resources + controlled environment + dynamics $\rightarrow$ organized outcome.

That is the architectural analogy, not a definition of computation.

5 Boundaries, environment, and dynamics

The comparison would still be incomplete if we treated either system as though it operated apart from its surroundings. The embryo develops through exchanges of matter and energy, while an engineered computer needs power, physical support, and conditions in which its operations can be carried out reliably. In both cases, the environment is part of the account, rather than merely a background setting.

This gives a more precise meaning to the boundary we found in the egg. Its role is not simply to exclude the outside world, but to allow some interactions while constraining others. A computer also depends on managing its surroundings, although its enclosures, power supply, cooling, and other protective arrangements serve different purposes from a shell and membranes. The resemblance concerns a requirement of the process, not an interchangeable mechanism.

Quantum computing offers a specialized example of this requirement: unwanted coupling to the environment can cause decoherence and errors, making environmental control particularly important. It illustrates the broader question without making an egg a quantum processor or an eggshell an error-correcting code.

We are therefore brought back to dynamics. Asking where information is located is not enough; we must also ask how it is maintained and transformed while the system interacts with its environment. The useful comparison is not between two static packages of information, but between physical processes that depend on the continuing interaction of several components.

Looking back, the egg was a useful starting point because it made those components visible together. Information does not produce an organized outcome by itself: it must participate in a physical process with machinery, resources, and suitable conditions. Computing provides another setting in which we can examine those roles explicitly.

6 From an egg to a question about computation

Our journey began with a very ordinary question: what is an egg? Asking where the life is in it led us to distinguish living machinery from the material that machinery consumes. That distinction opened up a further question: how can a tiny cellular system give rise to a vastly more complicated organism? We

then asked what is transmitted across generations: matter, information, or a mechanism for reconstructing organization? As resources and boundaries entered the picture, the question widened again: could this architecture also help us describe computation? Each step grew out of something the previous answer had left to explain.

The conclusion is not that an egg is secretly an engineered computer, or that every organized transformation should be called computation. It is more modest: considering development alongside computing helps us distinguish several roles that can otherwise be blurred together:

information machinery resources boundaries environment dynamics

An egg makes this architecture unusually visible because these components are packaged together within a single developmental system. An engineered computer carries out a different kind of task, but it too brings information into a physical process of transformation. Perhaps, then, the most useful question is not:

Is a living organism a computer?

That question is too broad to capture the distinctions we have found. A more useful question is:

What physical arrangements allow information to guide an organized transformation?

The egg and the computer illuminate different aspects of this question. Their comparison is a starting point for exploration, not a complete theory.

This brings me back to the interest described in the prefatory note. Information offers a way to distinguish roles within a physical process, while physics asks how those roles can actually be realized. The developing egg gives us a working picture of genetic information participating in a living process, without establishing that process as an engineered computation. It helps us ask more carefully what information can do, and what else must be present for it to do anything. An ordinary egg has given us a concrete route into those broader questions: **the physics of information becoming transformation.**

Side note: What if we stopped arguing about “veg” and “non-veg”?

The distinction we have been following also casts a different light on a familiar question about food: *Is an egg vegetarian?* Perhaps the usual categories are not the most precise way to explore what concerns us. Instead of asking whether something comes from an animal or a plant, imagine a hypothetical dietary rule defined at the level of biology: do not consume living cellular machinery itself. A narrower version would focus on reproductive cells and embryos, rather than on every living cell in a food.

Under that rule, the familiar boundary between vegetarian and non-vegetarian becomes less obvious. Consider the egg again. An egg is not simply a piece of chicken in the same sense as muscle tissue: it brings together the stored resources we have discussed and cellular material associated with the germinal disc. The question of what we are eating therefore becomes more specific. If our concern is the living biological machinery itself, what matters is not simply whether the object is called an egg, but whether we are consuming that cellular component.

Applying the same rule to a plant brings us back to the seed, which can likewise contain an embryo alongside stored resources and protective structures. This reveals an unusual symmetry. What if an egg containing its developmental cellular component were considered non-vegetarian, while an egg without that component were considered vegetarian? If that were our rule, what would happen when we applied it to a plant seed containing a living embryo? So perhaps, in this unusual biological sense, an egg would be called “non-veg” not because it comes from an animal, and a seed not because it comes from a plant, but because both can contain living machinery involved in the development of another organism. That is not what the word *vegetarian* normally means; it is a deliberately different criterion.

The apple pulp and the albumen of an egg, for example, become conceptually comparable in one respect: neither is itself the organism's developmental cellular machinery. They are biological material produced, accumulated or supplied as resources by a living organism. The seed, however, complicates the picture because it can contain the embryo itself. The distinction we are exploring is therefore not animal versus plant, but living cellular organization versus material produced or stored by that organization.

Under such a definition, one could imagine a very strict form of *vegetarianism* that applies the same rule to both kingdoms of life. It would probably be extremely difficult to formulate and sustain in practice, because biology does not divide food into such clean categories. Many foods contain mixtures of cellular and non-cellular material, and processing can alter whether cells remain viable. As a thought experiment, however, it gives us a more explicit criterion to examine. Instead of repeatedly asking *Is this veg or non-veg?*, we could ask: am I consuming the living biological machinery itself, or material that living machinery has produced, stored, or accumulated?

Could we turn this into an actual dietary category? One might combine two restrictions: avoid reproductive cells and embryos, and avoid animal tissue whose acquisition requires killing the animal. The second is an ethical criterion about how food is obtained, not a biological criterion about whether its cells are alive. Taken together, they suggest an unusual pattern: vegetative plant parts, fruits with seeds removed, and animal secretions such as milk, while excluding animal tissues, reproductive cells, and plant seeds or embryos. This is not equivalent to avoiding all living cells: the permitted plant foods are themselves cellular tissues. Nor does permitting milk settle the ethical questions surrounding its production. The category would therefore need explicit choices, rather than following automatically from a single distinction between life and non-life.

An illustrative menu might include potatoes and other tubers, selected vegetables, deseeded fruits, dairy, and carefully selected fats. But a list of permitted foods is not evidence of a nutritionally complete diet. Excluding seeds would also remove familiar staples such as grains, beans, lentils, and many nuts, including several common vegetarian protein sources. ^{ootnote}For ordinary balanced-diet guidance, see the USDA MyPlate discussion of protein foods and the NHS guide to vegetarian nutrition. Neither validates the hypothetical restrictions proposed here. Whether a particular menu supplies enough energy, protein, and essential nutrients would require assessment of its foods and quantities. Fortified foods or supplements might be needed, but should address identified gaps rather than be assumed to make any restrictive diet adequate. Anyone considering such a pattern in practice should seek advice from a registered dietitian.

These possibilities are a thought experiment, not a nutritionally validated dietary recommendation. Their value here is in exposing what our categories actually mean: are we concerned with animal origin, living cells, reproductive potential, or the consequences of obtaining the food? Those questions are related, but they are not interchangeable. This brings us back to the central theme of the essay. The egg mattered not merely as an egg, but as a seemingly ordinary object in which we could distinguish different layers: information, living machinery, resources, boundaries, and dynamics. The seed allowed us to recognize those layers again. Perhaps, then, the deeper question behind the apparently simple argument about food is the one we have been approaching all along: what is the difference between matter, life, information, and the machinery that organizes them?